Linear Systems Theory

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Module 2
Lecture 4

Math Preliminaries: Linear Algebra 3

Linear Maps / Linear Transformations

A map $f : U \to V$ between two vector spaces U (domain) and V (co-domain) over field $\mathbb{F}$ is linear if for $\mathbf{x}, \mathbf{y} \in U$ and $c, c_1, c_2 \in \mathbb{F}$, the following holds:

1. **Homogeneity**: $f(c\mathbf{x}) = cf(\mathbf{x})$
2. **Additivity**: $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$; $f(\mathbf{x}), f(\mathbf{y}) \in V$
3. **Superposition**: $f(c_1\mathbf{x} + c_2\mathbf{y}) = c_1 f(\mathbf{x}) + c_2 f(\mathbf{y})$

- In general, for $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \in U$ and $c_1, c_2, \ldots, c_n \in \mathbb{F}$

$$f(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \ldots + c_n\mathbf{u}_n) = c_1 f(\mathbf{u}_1) + c_2 f(\mathbf{u}_2) + \ldots + c_n f(\mathbf{u}_n)$$

Linear Maps/ Transformations: Examples

Following are examples of linear maps/transformations:

1. Identity transformation: The transformation $f : V \to V$, where $f(\mathbf{x}) = \mathbf{x} \; \forall \; \mathbf{x} \in V$
2. Zero transformation: The transformation $f : V \to V$ which maps each element of V to $\mathbf{0}$ i.e., $f(\mathbf{x}) = \mathbf{0} \; \forall \; \mathbf{x} \in V$
3. Multiplication by a scalar: The transformation $f : V \to V$, where $f(\mathbf{x}) = c\mathbf{x} \; \forall \; \mathbf{x} \in V, c \in \mathbb{R}$
4. Inner Product: The transformation $f : V \to \mathbb{R}$ such that $f(\mathbf{x}) = \mathbf{x}.\mathbf{z} \; \forall \; \mathbf{x} \in V$, where $\mathbf{z}$ is a fixed vector in the Euclidean space V
5. $f : \mathbb{R}^2 \to \mathbb{R}^3$ such that $f(\mathbf{x}) = f(x_1, x_2) = \begin{bmatrix} x_1 + 3x_2 \\ 4x_1 - x_2 \\ 2x_1 + x_2 \end{bmatrix}$

Image and Kernel of Linear Maps

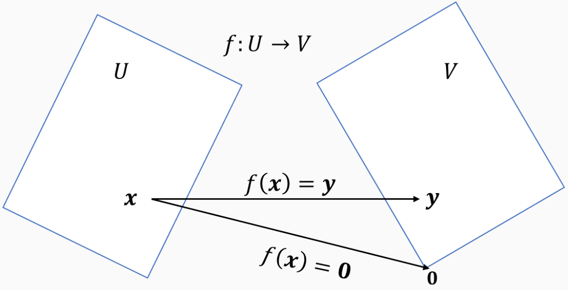
- Image of a linear map $f : U \to V$ is:

$$\mathbf{Im}(f) = \{\mathbf{y} \in V \mid \mathbf{y} = f(\mathbf{x}) \ \forall \ \mathbf{x} \in U\}$$

- $\mathbf{Im}(f)$ is a subspace of V as it satisfies the conditions for a subspace
- Kernel of a linear map $f : U \to V$ is:

$$\ker(f) = \{\mathbf{x} \mid f(\mathbf{x}) = \mathbf{0} \ \forall \ \mathbf{x} \in U\}$$

- $\ker(f)$ is a subspace of U as it satisfies the conditions for a subspace



System of Linear Equations as a Linear Map

- Consider a system of m linear equations in n unknowns:

$$\sum_{\mathrm{k}=1}^{n} a_{i\mathrm{k}} x_{\mathrm{k}} = b_i \text{ for } i = 1, \dots, m$$

  where $(x_1, x_2, \dots, x_n)$ are the n unknowns
- The coefficients a_{ik}'s can be written as a matrix $\mathbf{A}$ i.e., $\mathbf{Ax} = \mathbf{b}$
- Matrix $\mathbf{A}$ represents a linear transformation which maps $f : \mathbb{R}^n \to \mathbb{R}^m$ which maps vector $\mathbf{x} = (x_1, \dots, x_n)$ onto the vector $\mathbf{b} = (b_1, \dots, b_m)$
- The system has:
  - a unique solution if $\mathbf{b}$ is in the image of f and kernel of f contains only zero vector
  - many solutions if $\mathbf{b}$ is in the image of f and kernel of f is non-trivial
  - no solution if $\mathbf{b}$ is not in the image f

Linear Space of Transformations

- Let $\mathcal{L}(U, V)$ denote the set of all linear transformations of U into V over the field $\mathbb{F}$
- If $f : U \to V$ and $g : U \to V$ are two linear transformations in $\mathcal{L}(U, V)$, then we can define addition and scalar product with $\mathbf{x} \in U$ and $c \in \mathbb{F}$ as:

$$
\begin{aligned}
(f+g)(\mathbf{x}) &= f(\mathbf{x}) + g(\mathbf{x}) \\
(cf)(\mathbf{x}) &= cf(\mathbf{x})
\end{aligned}
$$

- It can be verified that $f + g$ and cf are also linear transformations in $\mathcal{L}(U, V)$
- Therefore, $\mathcal{L}(U, V)$ is a linear space by itself with zero transformation as the zero element of this space

Matrices as Linear Maps

Matrices as Linear Maps

- Consider a linear map $f : U \to V$ with U and V being vector spaces of dimensions n and m respectively
- Let the basis of U and V be $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ respectively
- Transformation of each of the basis vectors of U exists in V i.e., $f(\mathbf{u}_\mathrm{k}) \in V$
- Therefore, $f(\mathbf{u}_\mathrm{k})$ can be written as linear combination of basis vectors of V:

$$f(\mathbf{u}_\mathrm{k}) = \sum_{i=1}^{m} c_{i\mathrm{k}} \mathbf{v}_i$$

where $c_{1\mathrm{k}}, \ldots, c_{m\mathrm{k}}$ are the coordinates of $f(\mathbf{u}_\mathrm{k})$ w.r.t. the basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$

Matrices as Linear Maps

- The coordinates of $f(\mathbf{u}_k)$ can be written in a vector form and coordinates corresponding to transformations of all n basis vectors can be stacked as a matrix

$$\mathbf{A} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

- Every linear transformation f of an $n-$dimensional space U into an $m-$dimensional space V gives rise to an $m \times n$ matrix $\mathbf{A}$ whose columns consist of coordinates of $f(\mathbf{u}_1), \dots, f(\mathbf{u}_n)$ relative to the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$
- Therefore, a linear mapping $f : U \to V$ can be represented as a matrix-vector product:

$$f(\mathbf{x}) = \mathbf{A}\mathbf{x} \; ; \mathbf{x} \in U \text{ and } f(\mathbf{x}) \in V$$

Matrices as Linear Maps: Example

- Eg. Find the matrix representation of linear map $f(x_1, x_2) = \begin{bmatrix} x_1 + 3x_2 \\ 4x_1 - x_2 \\ 2x_1 + x_2 \end{bmatrix}$ considering standard basis for $\mathbb{R}^2$ and $\mathbb{R}^3$

$$f(1,0) = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} ; f(0,1) = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} ; \mathbf{A} = [f(1,0)\ f(0,1)] = \begin{bmatrix} 1 & 3 \\ 4 & -1 \\ 2 & 1 \end{bmatrix}$$

Exercise 1

For the above defined linear map, what is the matrix representation considering the basis $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ for $\mathbb{R}^2$?

- Eg. Given $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathbb{R}^2$. Find $f(\mathbf{x})$ in $\mathbb{R}^3$ via the linear map $f(x_1, x_2) = \begin{bmatrix} x_1 + 3x_2 \\ 4x_1 - x_2 \\ 2x_1 + x_2 \end{bmatrix}$ assuming standard basis.

  $f(\mathrm{x}) = \mathrm{Ax}$

Exercise 2

For the above defined linear map, what is $f(\mathbf{x})$ given $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$?

Overview

Summary: Mod 2 Lecture 4

- Linear Maps
- Image and Kernel
- Matrices as linear maps

Contents: Mod 2 Lecture 5

- Fundamental subspaces of a matrix
- Rank and Nullity
- Rank-Nullity Theorem
- Fundamental theorem of Linear Algebra